

Engineering and Backend Development

FastAPI

FastAPI is a Python framework used to build APIs. It is built on Starlette for web functionality and Pydantic for data validation. FastAPI supports asynchronous programming and automatically generates OpenAPI documentation.

FastAPI applications are commonly structured with routes, request models, response models, services, and database layers. Routes define HTTP endpoints such as GET, POST, PUT, and DELETE.

PostgreSQL

PostgreSQL is a relational database commonly used in backend applications. It supports transactions, indexes, foreign keys, and complex SQL queries.

Redis

Redis is an in-memory data store often used for caching, session storage, rate limiting, and temporary data.

Docker

Docker packages applications and their dependencies into containers. Containers provide consistent environments between development, testing, and production.

Kubernetes

Kubernetes is used to orchestrate containerized applications. It can manage deployments, scaling, service discovery, and container health.